



Technical article

Risk is the world's leading financial risk management magazine. **Risk's** Cutting Edge articles are a showcase for the latest thinking and research into derivatives tools and techniques, with comprehensive coverage of the most recent advances in areas such as option pricing and hedging, market risk, credit risk, swaps and Monte Carlo methods.

www.risk.net/technical

Copyright statement

This PDF is made available for personal use only. Under the following terms you are granted permission to view online, print out, or store on your computer a single copy, provided this is intended for the benefit of the individual user. Should you wish to use multiple copies of this article, please contact reprints@riskwaters.com

© **Risk Waters Group Ltd**, 2003. All rights reserved. No parts of this article may be reproduced, stored in or introduced into any retrieval system, or transmitted, in any form or by any means, electronic, mechanical, photocopying, recording or otherwise without the prior written permission of the copyright owners.

The essentials of the LMM

Using an innocuous approximation for forward rates, John Hull and Alan White develop an easy-to-implement version of the Libor market model

In recent years, Brace, Gatarek & Musiela (1997), Jamshidian (1997) and Miltersen, Sandmann & Sondermann (1997) have developed what has become known as the Libor market model (LMM). This is an important extension of the Heath, Jarrow & Morton (HJM, 1992) model. Its implementation is discussed in Hull & White (2000). While the HJM model describes the behaviour of instantaneous forward rates expressed with continuous compounding, the LMM describes the behaviour of the forward Libor rates underlying caps and floors, with the usual market conventions being used for compounding (that is, the compounding period equals the tenor of the rate). In essence, LMM is a “user-friendly” version of HJM. It is easier to use and easier to calibrate to market data.

The LMM models the forward rates underlying a cap. Suppose, for example, that the cap has a three-month tenor. The forward rates are for 3×6 months, 6×9 months, 9×12 months, and so on. The initial values of the forward rates are known. The LMM describes the stochastic process governing the changes in these forward rates.

The notation we use is as follows. We consider a cap with resets at times $t_1, t_2, \dots, t_n$ and a final payment date t_{n+1} . We define $t_0 = 0$, $\delta_i = t_{i+1} - t_i$ ($0 \leq i \leq n$), $F_i(t)$ as the forward rate observed at time t for the period (t_i, t_{i+1}) expressed with a compounding period of δ_i ($t \leq t_i$), $m(t)$ as the index for the next reset date at time t (which means that $m(t)$ is the smallest integer such that $t \leq t_{m(t)}$), p as the number of factors and $\zeta_{i,q}(t)$ as the q th component of the volatility of $F_i(t)$ ($1 \leq q \leq p$).

The LMM is:

$$dF_i = \dots dt + \sum_{q=1}^p \zeta_{i,q}(t) F_i(t) dz_q \quad (1)$$

where the dz_q s are independent Wiener processes. The stochastic process for each forward rate has p independent factors. The q th component of the volatility of the i th forward rate is $\zeta_{i,q}(t)$. We assume this is a function only of time. This leads to all forward rates being lognormal at all times – a result that is consistent with the standard market model for valuing caplets.

The drift of each forward rate depends on the numeraire. We choose as the numeraire a discretely reinvested money market account. This is created by starting with \$1, investing it for the first accrual period, reinvesting it for the second accrual period, and so on. The coefficient of dt is then:

$$\sum_{j=m(t)}^i \frac{\delta_j F_j(t) \sum_{q=1}^p \zeta_{j,q}(t) \zeta_{i,q}(t) F_i(t)}{1 + \delta_j F_j(t)}$$

The numeraire specified above is a convenient one to use. It has the advantage that discounting an expected cashflow at time t_{i+1} to time t_i is straightforward. The correct discount rate is the δ_i -period Libor rate observed at time t_i .

To simplify the model further, we assume that $\zeta_{i,q}(t)$ is a function only of the number of whole accrual periods between time t and the maturity

of the forward contract, time t_i . This means that it is a step function of time. The variable $\lambda_{j,q}$ is defined as the q th component of the volatility when there are j whole accrual periods so that $\zeta_{i,q}(t) = \lambda_{i-m(t),q}$. To illustrate how the model works, consider the 21×24 forward rate. This is the seventh forward rate, so $i = 7$. For the first three months ($0 \leq t \leq 0.25$), $m(t) = 1$, so that the q th component of the forward rate volatility is $\lambda_{6,q}$; for the next three months, $m(t) = 2$ so that the q th component of the forward rate volatility is $\lambda_{5,q}$; and so on.

When calculating the coefficient of dt in equation (1), we make the approximation that the forward rate calculated during an accrual period equals the forward rate at the beginning of the accrual period: $F_i(t) = F_i(t_i)$ for $t_i < t < t_{i+1}$. This approximation is tested in Hull & White (2000) and found to be innocuous. It makes very little difference to the way the model prices caps. However, its advantage is that it makes the implementation of the model much easier. In particular, it leads to the following equation for simulating forward rates:

$$F_i(t_{k+1}) = F_i(t_k) \exp \left[\left(\sum_{j=k+1}^i \frac{\delta_j F_j(t_k) \sum_{q=1}^p \lambda_{j-k-1,q} \lambda_{i-k-1,q}}{1 + \delta_j F_j(t_k)} - \sum_{q=1}^p \frac{\lambda_{i-k-1,q}^2}{2} \right) \delta_k + \sum_{q=1}^p \lambda_{i-k-1,q} \epsilon_q \sqrt{\delta_k} \right] \quad (2)$$

where ϵ_q are independent samples from standard normal distributions.

As mentioned earlier, our choice of the numeraire allows us to jump from one reset date to the previous one when discounting. The drift approximation just mentioned allows us to jump from one reset date to the next when moving forward in a Monte Carlo simulation. It makes sense for us to set up the model this way because, for most interest rate derivatives, nothing happens between reset dates. There are no cashflows and no early exercise opportunities.

Cap and European swap option pricing

The volatility used to price a caplet in the standard market model can be calculated from the λ s. Define the total volatility of a forward rate when there are j whole accrual periods until maturity as Λ_j . The relationship between the Λ s and λ s is:

$$\Lambda_j^2 = \sum_{q=1}^p \lambda_{j,q}^2$$

The volatility used in Black's (1976) model to price a cap with resets at times $t_0, t_1, t_2, \dots, t_n$ is:

$$\frac{1}{t_n} \sum_{j=1}^n \Lambda_{n-j}^2 \delta_{j-1} \quad (3)$$

Note that the pricing of vanilla caplets, and therefore vanilla caps, depends only on the Λ s. It does not depend on how that volatility is split into its components. Vanilla caplet pricing is therefore independent of the number of factors. Interestingly, the same is not true of exotic caps. Hull (2000) considers ratchet caps, sticky caps and flexi caps, and shows that their prices increase as the number of factors increase (with the Λ_j being held fixed). This is because the values of the options embedded in these products become greater as the correlations between the movements in different forward rates decline.

European-style swap options can be priced under the LMM using an analytic approximation. Consider a swap with reset dates at times $t_n, t_{n+1}, \dots, t_N$ and a final payment date t_{N+1} . In Hull & White (2000), we use Itô's lemma to calculate the volatility of the swap rate at time t . When calculating this volatility, we make the approximation that all forward rates remain constant at their initial values: $F_i(t) = F_i(0)$. This approximation leads to the following estimates of the swap option volatility used in Black's (1976) model:

$$\sqrt{\frac{1}{t_n} \sum_{j=0}^{n-1} \left\{ \delta_j \sum_{q=1}^p \left[\sum_{k=n}^N \frac{\delta_k \lambda_{k-j-1,q} F_k(0) \gamma_k(0)}{1 + \delta_k F_k(0)} \right]^2 \right\}} \quad (4)$$

where:

$$\gamma_k(t) = \frac{\prod_{j=n}^N [1 + \delta_j F_j(t)]}{\prod_{j=n}^N [1 + \delta_j F_j(0)] - 1} - \frac{\sum_{i=n}^{k-1} \delta_i \prod_{j=i+1}^N [1 + \delta_j F_j(t)]}{\sum_{i=n}^N \delta_i \prod_{j=i+1}^N [1 + \delta_j F_j(0)]} \quad (5)$$

This approximation is tested in Hull & White (2000) and, like our earlier approximation, is found to be innocuous. European-style swap option prices given by a full simulation of the LMM are remarkably close to those obtained from using equation (4) for the parameter values normally en-

countered in financial markets.

Equation (4) is the volatility that should be substituted into the standard market model for pricing European-style swap options. It is derived by assuming that the tenor of the swap underlying the swap option is the same as the tenor of the caps that are considered. This is not always the case. In the US, for example, the tenor of the swaps underlying the most actively traded swap options is six months while the tenor of the most actively traded caps is three months. Hull & White (2000) show how the analysis leading to equation (4) can be extended to accommodate the situation where the swap tenor is a multiple of the cap tenor.

Calibration

Like Rebonato (1999), we favour using market data to determine the Λ s and historical data to determine how the Λ s should be split into λ s. For any given set of Λ s, the λ s are determined from the Λ s using the output from a principal components analysis. The principal components analysis model is:

$$\Delta F_j = \sum_{q=1}^N \alpha_{j,q} x_q$$

where:

$$\sum_{j=1}^N \alpha_{j,q_1} \alpha_{j,q_2}$$

equals one when $q_1 = q_2$ and zero when $q_1 \neq q_2$. Define s_q as the standard deviation of the random variable x_q . We set:

$$\lambda_{j,q} = \frac{\Lambda_j s_q \alpha_{j,q}}{\sqrt{\sum_{q=1}^p s_q^2 \alpha_{j,q}^2}}$$

In theory, the Λ s can be determined directly from caplet volatilities using a bootstrap procedure. Using equation (3), the first caplet volatility determines Λ_0 ; the next caplet volatility determines Λ_1 ; and so on. In practice, this procedure quite often leads to “spiky” Λ s and sometimes gives rise to a negative value for one or more of the Λ_j^2 . A more robust procedure is to use the least squares criterion to determine the Λ s from an appropriate set of calibrating instruments (usually caps and European-style swap options). This involves minimising an expression of the form:

$$\sum_{k=1}^K (u_k - v_k)^2 + P$$

where K is the number of calibrating instruments, u_k is the market price of the k th calibrating instrument and v_k is its price given by the model. The term P contains penalties relating to the gradient and curvature of Λ_j as a function of j and is chosen to ensure that Λ is reasonably smooth.

The extended LMM

The volatility skew for equities has been well documented by authors such as Jackwerth & Rubinstein (1996). As pointed out by Andersen and Andreasen (1997), caps and floors exhibit a similar volatility skew to equities. The volatility is high for low strike prices. It decreases as the strike price increases. When the strike price becomes very high, the volatility may start increasing again, but the overall pattern is more of a skew than a smile. Some brokers do provide quotes for caps that are not at-the-money, but there is very little data on pricing non-at-the-money European-style swap options. It is, therefore, of interest to investigate how cap volatility skews can be converted to swap option volatility skews.

As shown by Andersen & Andreasen (1997), the LMM can be extended to incorporate volatility skews by allowing the forward rates to follow a constant elasticity of variance (CEV) model. This is the model in equation (1) with the $F_i(t)$ on the right-hand side being replaced by $F_i(t)^\alpha$ for some constant α . Hull & White (2000) show that, as an approximation, swap rates follow a similar CEV process to forward rates. Caps can be priced exactly in terms of the cumulative non-central chi-square distribution. A similar analysis to that leading to equation (4) allows swap options also to be priced in terms of the cumulative non-central chi-square distribution.

This model, known as the extended LMM, allows cap volatility skews to be converted to swap option volatility skews. The calculations are very fast. All they require is that the CEV parameter, α , be estimated periodically using a least squares approach from the at-the-money and non-at-the-money cap volatilities provided by brokers.

Valuing Bermudan swaptions

The LMM must be implemented using Monte Carlo simulation and does not naturally lend itself to the valuation of Bermudan options. Fortunately, a number of researchers, such as Andersen (1999), Broadie & Glasserman (1997) and Longstaff & Schwartz (1998), have found creative ways of valuing these types of options. Andersen's approach may be the easiest to apply. To implement a version of his approach that he finds works well, we proceed as follows. We assume that the early exercise decision depends only on the option's intrinsic value. Specifically, we assume that we exercise at time t_i if the intrinsic value is greater than some level H_i .

REFERENCES

Andersen L, 1999

A simple approach to the pricing of Bermudan swaptions in the multifactor Libor market model
Journal of Computational Finance 3(2), winter, pages 5–32

Andersen L and J Andreasen, 1997

Volatility skews and extensions of the Libor market model
Working paper, Gen Re Financial Products, forthcoming in Applied Mathematical Finance

Brace A, D Gatarek and M Musiela, 1997

The market model of interest rate dynamics
Mathematical Finance 7(2), pages 127–155

Black F, 1976

The pricing of commodity contracts
Journal of Financial Economics 3, March, pages 167–179

Broadie M and P Glasserman, 1997

A stochastic mesh method for pricing high dimensional American options
Working paper, Columbia University, New York

Heath D, R Jarrow and A Morton, 1992

Bond pricing and the term structure of interest rates: a new methodology
Econometrica 60(1), pages 77–105

Hull J, 2000

Options, futures, and other derivatives
Prentice Hall, fourth edition

Hull J and A White, 2000

Forward rate volatilities, swap rate volatilities, and the implementation of the Libor market model
Journal of Fixed Income 10(2), September, pages 46–62

Jackwerth J and M Rubinstein, 1996

Recovering probability distributions from option prices
Journal of Finance 51, December, pages 1,611–1,631

Jamshidian F, 1997

The Libor and swap market models
Finance and Stochastics 1, pages 293–330

Longstaff F and E Schwartz, 1998

Valuing American options by simulation: a least squares approach
Working paper 25-98, Andersen School at the University of California at Los Angeles

Miltersen K, K Sandmann and D Sondermann, 1997

Closed form solutions for term structure derivatives with lognormal interest rates
Journal of Finance 52(1), March, pages 409–430

Rebonato R, 1999

On the simultaneous calibration of multifactor lognormal interest rate models to Black volatilities and to the correlation matrix
Journal of Computational Finance 2(4), summer, pages 5–27

To determine the H_i , we carry out Monte Carlo simulations in the usual way. For each simulation trial, we store the relevant intrinsic values and the relevant discount factors. We work back choosing the H_i to maximise the value of the American-style swap option. Once the H_i s have been chosen in this way, we carry out a fresh Monte Carlo simulation to determine the option price. ■

John Hull and Alan White are finance professors in the Joseph L Rotman School of Management at the University of Toronto

Comments on this article can be posted on the technical discussion forum on the Risk website at <http://www.risk.net>